MBG 1201 Introduction to Modern Biology A

Name Surname: Signature:

Please encircle the letter of the correct answer. Each question is 4 pts. Select 25 questions out of 30. Each extra question you answer correctly will be considered as 2 pts and be added to your total points from other exams if required. You have 70 minutes. Good luck.

1. An action potential begins when_____ .

A) sodium–potassium pumps stop operating

B) voltage-gated potassium gates open

C) voltage-gated sodium gates close

D) a neuron reaches threshold potential

2. All mutations _____.

A) result from radiation

B) lead to evolution

C) are caused by DNA damage

D) change the DNA sequence

3. In operon models, the function of the promoter is to

A) code for the repressor protein.

B) bind with RNA polymerase.

C) bind to the repressor.

D) code for the regulator gene.

4. Which of the following tools of DNA technology is incorrectly paired with its use?

A) electrophoresis—separation of DNA fragments

B) DNA ligase—cutting DNA, creating sticky ends of restriction fragments

C) DNA polymerase—polymerase chain reaction to amplify sections of DNA

D) reverse transcriptase—production of cDNA from Mrna

5. Which term describes the space between a neuron and its target cell?

A) Post synaptic membrane

B) Synaptic cleft

C) Dendritic spine

D) Axon terminal

6. Energy that drives the attachment of a nucleotide to the end of a growing strand of DNA comes from _____.

A) the nucleotide

B) DNA polymerase

C) phosphate-group transfers from ATP

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7. If the sequence of DNA on the template strand of a gene is AAA, the mRNA codon produced by transcription will be _____ and will specify the amino acid _____ . Use the codon table below:

Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
GCA	AGA						GGA			UUA				CCA	AGC	ACA			GUA	
GCC	AGG						GGC		AUA	UUG				CCC	AGU	ACC			GUC	UAA
GCG	CGA	GAC	AAC	UGC	GAA	CAA	GGG	CAC	AUC	CUA	AAA		UUC	CCG	UCA	ACG		UAC	GUG	UAG
GCU	CGC	GAU	AAU	UGU	GAG	CAG	GGU	CAU	AUU	CUC	AAG	AUG	UUU	CCU	UCC	ACU	UGG	UAU	GUU	UGA
	CGG									CUG					UCG					
	CGU									CUU					UCU					

A) AAA, lysine B) AAA, phenylalanine C) TTT, arginine

D) UUU, phenylalanine E) TTT, lysine

8. Suppose an investigator at the scene of a murder finds a pair of gloves. Back at the lab, she discovers small fragments of skin and hair on the gloves. What fast test might she do with this evidence to help solve the case?

A) Remove the nucleus of a skin cell and place it in an egg cell

B) Extract DNA and determine the genotypes at multiple STR short tandem repeat) sites

C) Extract DNA and sequence the genome

D) Use DNA probes to search for specific gene sequences

9. Independent assortment means _____.

A) alleles at one locus assort into different gametes

B) alleles at different loci assort into gametes independently of each other

C) the assortment of genes on homologous chromosomes varies independently

10. A tumor suppressor gene

A) inhibits cell division. B) opposes oncogenes. C) prevents cancer.

D) is subject to mutations. E) All of these are correct.

11. In the list of four terms below, which term is the second most inclusive?

A) Genome B) Chromosome C) Allele D) Gene

12. Translational control of gene expression occurs within the

A) nucleus B) cytoplasm. C) nucleolus. D) mitochondria.

13. How does X inactivation contribute to a person's phenotype?

A) It controls the number and kind of genes inherited on an X chromosome.

B) It determines which X chromosome is expressed in a male.

C) It allows for the expression of either the maternal or the paternal X in different cells.

D) It enhances the expression of Y-linked genes in males.

14. Crossing over mixes up _____.

A) chromosomes B) zygotes C) alleles D) gametes

15. Why are action potentials usually conducted in one direction?

A) Ions can flow along the axon in only one direction.

B) The brief refractory period prevents reopening of voltage-gated Na+ channels.

C) The axon hillock has a higher membrane potential than the terminals of the axon.

D) Voltage-gated channels for both Na+

16. Choose the statement that is <u>incorrect</u>.

A) A son can inherit a recessive allele on an X chromosome from either parent.

B) An individual may inherit three or more of each type of chromosome characteristic of the species, a condition called polyploidy.

C) A female child inherits one X chromosome from her mother and one from her father.

D) Pedigree analysis can be used to determine a future child's chance of being born with achondroplasia.

17. The chromosome number _____.

A) refers to a particular chromosome in a cell

B) is a characteristic feature of a species

C) is the number of autosomes in cells of a given type

D) is the same in all species

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18. If the sequence of bases in the coding strand of a DNA molecule is TAGC, then the sequence of bases in the mRNA will be.

A) AUCG. B) TAGC. C) UAGC. D) ATCG.

19. Which component is directly involved in translation?

A) RNA polymerase B) ribosome C) spliceosome D) DNA

20. In DNA technology, the term vector can refer to

A) the enzyme that cuts DNA into restriction fragments.

B) the sticky end of a DNA fragment.

C) a SNP (single nucleotide polymorphisms) marker.

D) a plasmid used to transfer DNA into a living cell.

21. Dogs have a diploid chromosome number of 78. How many chromosomes do their gametes have?

A) 39 B) 156 C) 78 D) 234

22. Neurotransmitters are released by_____ .

A) axon terminals B) dendrites C) a neuron cell body D) glial cells

23. The portion of the mRNA transcript that gets removed during RNA processing is the

A) exons. B) introns. C) poly-A tails. D) 5' caps.

24. If a restriction enzyme cuts between the G and the A whenever it encounters the sequence GAATTC, how many fragments will be produced when the enzyme is digested with DNA with the following sequence?

TGAGAATTCAACTGAATTCAAATTCGAATTCTTAGC

A) Two B) Four C) Three D) Five

25. Most human axons have a myelin sheath that _____.

A) prevents action potentials from moving backward

B) insulates them and increases the speed of action potential conduction

C) consists of specialized interneurons

D) increases the size of action potentials

26. The phrase "5 prime to 3 prime" refers to the _____.

A) timing of DNA replication B) directionality of DNA synthesis

C) number of phosphate groups

27. Which of the following regulate(s) gene expression in the eukaryotic nucleus?

A) posttranslational control B) transcriptional control C) translational control

D) posttranscriptional control E) Both b and d are correct.

28. Which of the following is/are not required for DNA replication to occur?

A) DNA polymerase B) primers C) nucleotides

D) helicase E) template DNA F) all are required

29. An organism's observable traits constitute its _____.

A) phenotype B) genotype C) variation D) pedigree

30. A heterozygous individual has _____for a trait being studied.

A) the same allele on both homologous chromosomes

B) two different alleles of a gene C) a haploid condition, in genetic terms